

2.4 Example: AND gate vs OR gate

Consider a hypothetical circuit of three components A, B, and C that are connected with an AND or an OR gate. They are embedded in a much larger network, and of the three just C is connected to the output. We run an experiment where we patch all components using the denoising or noising technique.

AND circuit: $C = A \text{ AND } B$

- Denoising (clean $\rightarrow$ corrupt patching): Denoising either A or B has no effect on the output, only denoising C restores the output. This is because denoising A still leaves B at the corrupted (incorrect) baseline, and vice versa. *Denoising found only one of the circuit components.*
- Noising (corrupt $\rightarrow$ clean patching): Noising either A or B has an effect, as well as noising C.

Noising works better in this case, as it finds all circuit components in the first pass.

OR circuit: $C = A \text{ OR } B$

- Denoising (clean $\rightarrow$ corrupt patching): Denoising either A or B has an effect, as well as denoising C.
- Noising (corrupt $\rightarrow$ clean patching): Noising either A or B has no effect on the output, only denoising C restores the output. This is because noising A still leaves B at the clean (correct) baseline, and vice versa. *Denoising found only one of the circuit components.*

Denoising works better in this case, as it finds all circuit components in the first pass. These AND and OR structures can appear in real-world transformers as serial-dependent components (e.g. a later attention head depending on an earlier one) or parallel components (such as redundant backup attention heads).

2.5 Comparison to ablations and other patching techniques

There are activation patching techniques based on a single prompt. The original Causal Tracing (ROME, Meng et al., 2022) falls into this category, and also zero- and mean-ablation can be seen as patching techniques.

1. **Zero ablation:** Overwrite (“ablate”) the targeted activations with zeros and observe ablating which component breaks the model behaviour.
2. **Mean ablation:** Same as above but overwrite targeted activations with their dataset mean value rather than zero. This is slightly more principled than zero ablating since there is no special meaning to activations being zero.
3. **Gaussian noise patching** (also called Causal Tracing*): This is a clean $\rightarrow$ corrupt patching variant that uses as its corrupt run input the embeddings of the clean prompt with added Gaussian noise. The idea is to thereby automatically generate the corrupted “prompt”. It was originally used in ROME (called Causal Tracing there) but has not been used much recently, especially because the corruption can sometimes be ineffective.³

³The success of Gaussian noise corruption is highly sensitive to the noise level. Zhang and Nanda (2023) that if the noise level is just slightly lower than used in ROME, the model can recover the correct completion despite the corruption.

Technique	Source (saved)	Source run input	Destination / Base (over-written)	Destination / Base run input	Observation
Clean $\rightarrow$ corrupted (Denoising, Causal Tracing ²)	First run activations (clean)	Clean tokens	Second run activations (corrupted)	Corrupt tokens	What restores behaviour
Corrupted $\rightarrow$ clean (Noising, Resample Ablation)	First run activations (corrupted)	Corrupt tokens	Second run activations (clean)	Clean tokens	What breaks behaviour
Zero ablation	Zero activations	N/A	Clean run activations	Clean tokens	What breaks behaviour
Mean ablation	Dataset mean activations	N/A	Clean run activations	Clean tokens	What breaks behaviour
Gaussian Noise patching (Causal Tracing ²)	First run activations (clean)	Clean tokens	Second run activations (corrupted from modified clean input)	Clean token embedding + Gaussian noise	What restores behaviour

Generally we recommend corrupted-prompt-based techniques, noising and denoising. Their advantage is that one can run very precise experiments, editing some features while controlling for others. They allow us to trace *the difference between clean and corrupted prompt*. To illustrate this consider the prompts “Angela Merkel is the leader of” $\rightarrow$ “Germany” vs “Joe Biden is the leader of” $\rightarrow$ “America”. Patching will find components that deal with Angela Merkel vs Joe Biden, but not components that would be indifferent to this change, such as the “answer is a country circuit” or the “political leader circuit”. A secondary advantage of noising and denoising is that they tend to bring the model less out-of-distribution than ablation techniques (as pointed out in Chan et al. (2022), as well as in e.g. Hase, Xie, and Bansal, 2021)

2.6 Choosing corrupted prompts

Having a corrupted prompt is great because it can tell us what model components care about, but also a possible pitfall if we don’t notice what our prompts trace and don’t trace. We give some examples for the Indirect Object Identification (IOI, Wang et al., 2022) demo sentence “John and Mary went to the store. John gave a bottle of milk to”. Different corruptions which highlight different properties the model might care about include:

Corruption	Example	Property traced in model
None (Clean)	John and Mary ... John ...	
Replace the value of one or multiple names, without changing the grammatical structure	John and Alice ... John ... Alice and Mary ... Alice ... Alice and Bob ... Alice ...	Where the model represents the name values
Change which name is direct and indirect object without changing the names or positions	John and Mary ... Mary ...	The value and position of the indirect object
Change the position of the names without changing which one is subject and indirect object	Mary and John ... Mary ...	The value, but not the position of the indirect object (position is fixed)
Change a name to break the behaviour	John and Mary ... Alice ... Alice and Mary ... John ...	Specifics about IOI setting (e.g. that a name is duplicated at all)
Change all the names	Alice and Bob ... Charlie ...	Finding and confirming all relevant components

What kind of prompt should you choose? No matter which you choose, keep in mind what properties your prompt does and does not change, and take this into account when interpreting patching results. As a rule of thumb you want to choose small (narrow) variations for exploratory patching, this will help you narrow down what each component is tracking. Choosing a narrow prompt distribution also helps increase the (typically low) sensitivity of denoising, and decrease the (typically high) sensitivity of noising. For confirmatory patching you need to choose a wide distribution of prompts that varies all variables of the hypothesised circuit. Then you can noise (corrupt $\rightarrow$ clean patch) all non-circuit components, and check that the model still shows the behaviour in question.

²Causal Tracing has been used to describe ROME-style Gaussian noise patching in particular, but also to describe clean $\rightarrow$ corrupted patching in general. We recommend avoiding the name to avoid confusion.

3 How do you interpret patching evidence?

In the previous section we said that denoising (clean $\rightarrow$ corrupt patching) tests whether the patched activations are **sufficient to restore** model behaviour. And noising (corrupt $\rightarrow$ clean patching) tests whether the patched activations are **necessary to maintain** model behaviour. These two are usually not complements of each other, nor does one imply the other. In this section we will walk through a made-up example experiment.

3.1 Walkthrough a stylized example

Consider the hypothetical “Nobel Peace Prize” circuit. The model correctly completes “Nobel Peace” with “Prize”, using the following circuit:

- Attention head L0H0 is a “Previous Token Head” and copies the embedding of “Nobel” to the position of “Peace”
- Neuron L1N42 maps the mix of Nobel and Peace embeddings to the Prize logit
- Everything else doesn’t matter (of course a real circuit is typically much messier)

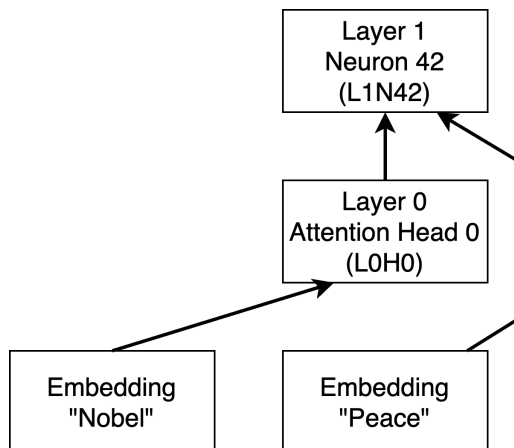


Figure 1: Toy “Nobel Peace Prize” circuit

Now let us run the standard patching examples, take a distribution of random English words for the corrupted prompt. We would find

- Noising (corrupt $\rightarrow$ clean patching) suggests that the outputs of head L0H0, the output of neuron L1N42, and the embeddings (Nobel & Peace) are all necessary components.
- Denoising (clean $\rightarrow$ corrupt patching) suggests that the output of neuron L1N42 is sufficient to restore the circuit.

What happened here? Denoising finds only the neuron output L1N42, because the other two components individually are not sufficient to restore the circuit behaviour! We’re dealing with an AND circuit between the attention head output and the “Peace” embedding. Noising finds all three components here.

Nonetheless denoising L1N42 alone restored the model behaviour. This is a crucial intuition to keep in mind about denoising: If you patch component A in layer N, it has seen clean versions of every component in layers 0 to N-1. If there’s an important component B in layer N-1 that is mediated by component A, the model can be restored without denoising B.

Patching experiments are sensitive to what precisely are the changes between the corrupt and clean prompt. If we created two additional corrupt distributions where we replace only either “Nobel” or “Peace” with a random word (i.e. distributions “X Peace” and “Nobel Y”) we could narrow down which component depends on which input.